



Osnovne operacije

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Digitalni zapis slike

- Slika sivih razina



```
A=imread('lena.jpg');  
size(A)
```

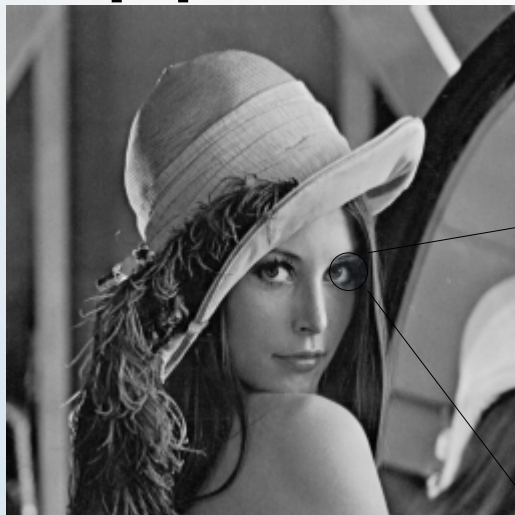
ans =

256 256

175 177 178 168 177
172 173 173 169 169 170
163 168 175 176 171 169 172
163 172 177 179 175 171 174
158 175 175 181 177 174 177
173 177 182 185 182 177
175 187 187 192 187

Tipovi podataka (u Matlabu)

- Tip podataka ***uint8***
 - 8 bitni zapis – raspon vrijednosti [0, 255]
 - Što sa operacijama koje za rezultat imaju vrijednost izvan raspona [0,255] ?
- Tip podataka ***double***



$[0, 255] \longrightarrow [0, 1]$

0.6863 0.6941 0.6980 0.6627
0.6745 0.6784 0.6784 0.6627
0.6392 0.6588 0.6863 0.6902 0.6706 0.6627
0.6392 0.6745 0.6941 0.7020 0.6863 0.6745
0.6196 0.6863 0.6863 0.7098 0.6941 0.6863
0.6775 0.6784 0.6941 0.7137 0.7255 0.6863
0.6863 0.7333 0.7333 0.7500

RGB zapis



```
R=A(:, :, 1);  
G=A(:, :, 2);  
B=A(:, :, 3);
```

```
A=imread('lenacolor.jpg');  
size(A)
```

ans =

256 256 3

237	236	235	234	237	237	237	237	237	237
237	238	237	237	233	243	238	240	240	240
234	167	173	175	174	172	171	176	176	176
235	164	171	175	175	171	170	181	175	170
234	161	154	158	160	159	157	156	158	158
237	161	152	157	160	160	157	155	166	160
239	156	152	160	164	164	161	158	156	156
237	157	150	161	166	166	162	159	162	162
237	164	159	165	166	166	166	166	166	166

Binarne slike



0 – crni pikseli

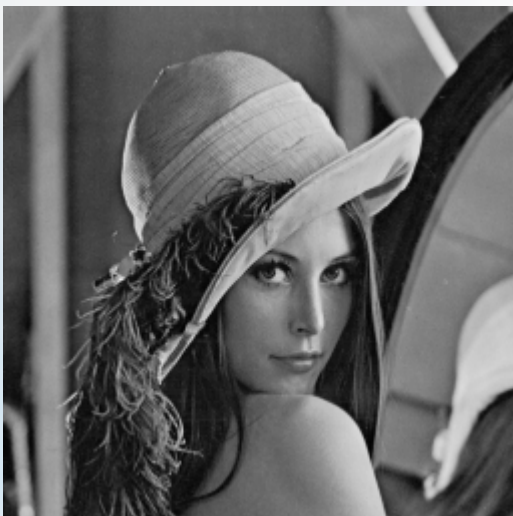
1 – bijeli pikseli

				0	0	1	1			
		0	0	0	1	0	0	0		
	0	0	0	1	0	0	0	0	0	
0	0	0	0	0	0	1	1	1	0	0
0	0	0	1	1	1	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	1	1	1	1
	0	0	0	0	0	1	0	0	0	
	0	0	0	0	0	0	0	1	0	
		0	0	0	0	0	0	1		

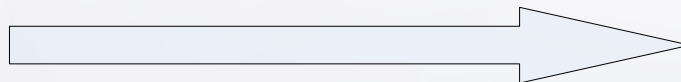
Region of Interest (ROI)

- Često nas zanima manji dio slike
 - Izdvajanje (uvećanje, zasebni prikaz)
 - Operacije na manjem dijelu ulazne slike (ušteta računalnih resursa)

```
A=imread('lena.jpg');
```



```
S=A(120:150, 100:180);
```



```
size(S)
```

```
ans =
```

```
31
```

```
81
```

Operacije na digitalnoj slici

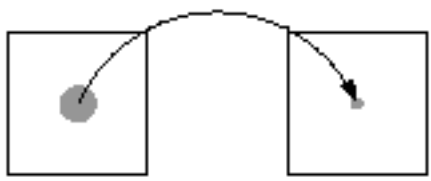
- **Operacije zasnovane na točki**

- Izlazna vrijednost na osnovu vrijednosti piksela ulazne slike na istoj koordinati



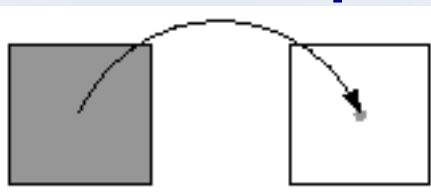
- **Lokalne operacije**

- Izlazna vrijednost temeljem ulazne na istoj koordinati i vrijednosti susjednih elemenata



- **Globalne operacije**

- Izlazna vrijednost ovisi o svim pikselima ulazne slike



Lookup table (LUT)

- Široka primjena u računalnoj znanosti
 - Korisiti se za zamjenu složenijih računskih operacija jednostavnom operacijom indeksiranja
 - Transformacija ulaznih podataka u izlazne vrijednosti
- **Struktura podataka sa izlaznom vrijednosti za sve moguće ulazne vrijednosti**
 - U formi jedan na jedan ili u obliku raspona vrijednosti
 - Ulazna vrijednost indeksira podatak u tablici transformacija
 - Efikasnost implementacije ovisi o veličini LUT tablice, memoriji, algoritmu pristupa kešu, ...

LUT - primjer 1

```
LT=uint8(255:-1:0);  
LA=LT(A);
```



LUT - primjer 1

```
LT=uint8(255:-1:0);  
LA=LT(A);
```



LUT - primjer 2

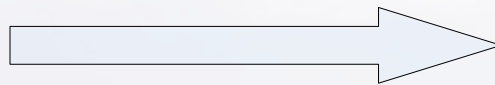
```
LT=uint8(255:-1:0);  
LA=LT(A);
```



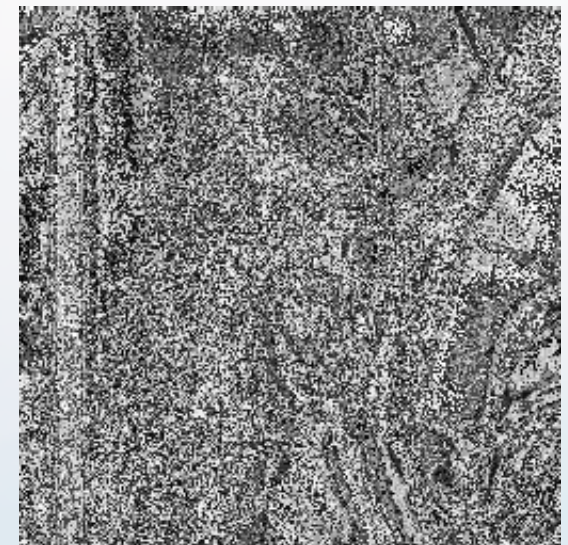
```
RT=uint8(rand(256,1)*255);  
RA=RT(A);
```

LUT - primjer 2

```
LT=uint8(255:-1:0);  
LA=LT(A);
```



```
RT=uint8(rand(256,1)*255);  
RA=RT(A);
```



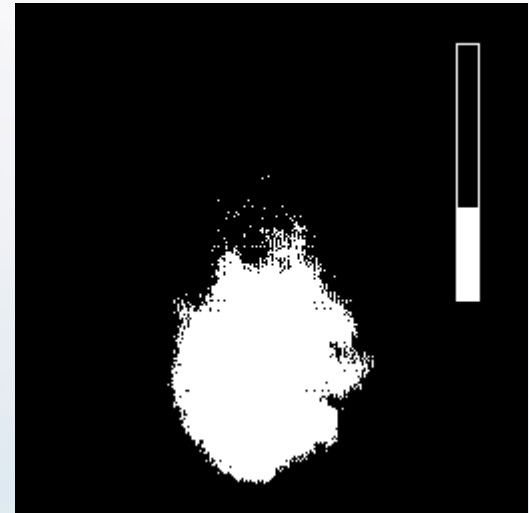
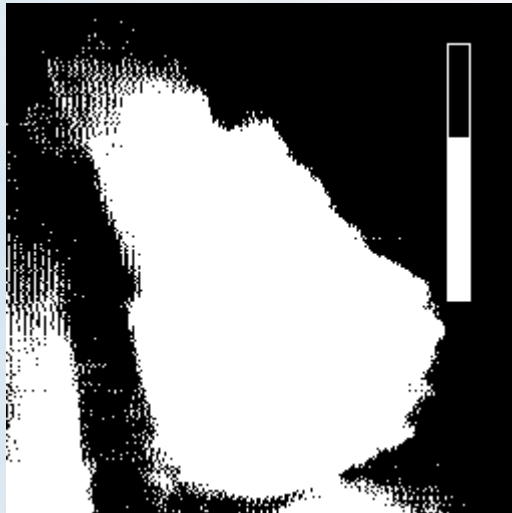
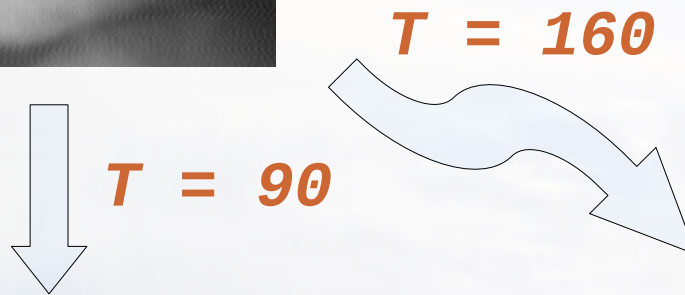
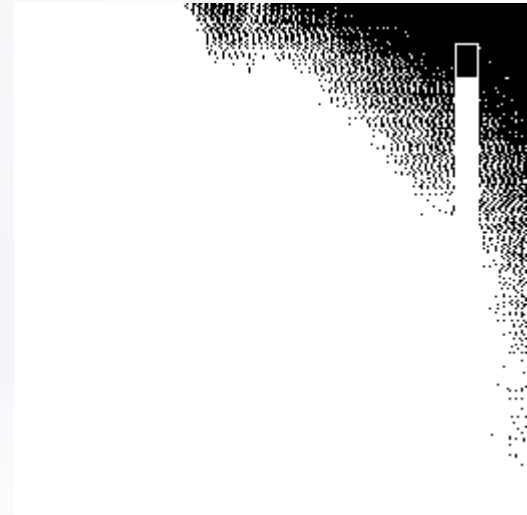
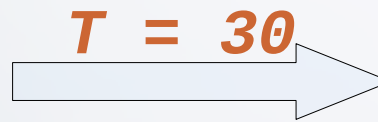
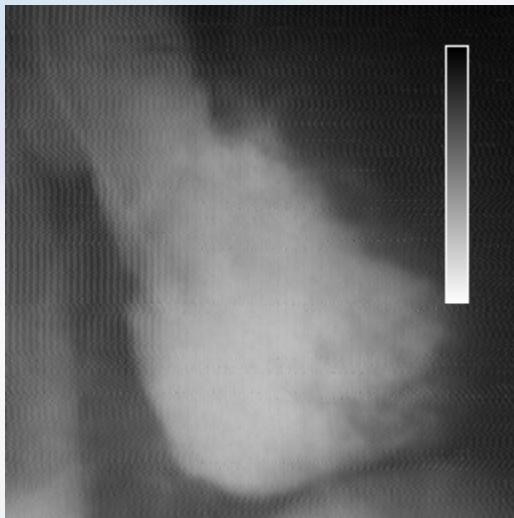
Amplitudna segmentacija (Thresholding)

- Podjela slike na dvije rejije postavljanjem praga

$$g(x, y) = \begin{cases} 0 & \text{if } f(x, y) < T \\ 1 & \text{otherwise} \end{cases}$$

- Korisna kada amplitudne značajke dovoljno precizno definiraju granicu dvije regije

Kako pdabrati iznos praga?



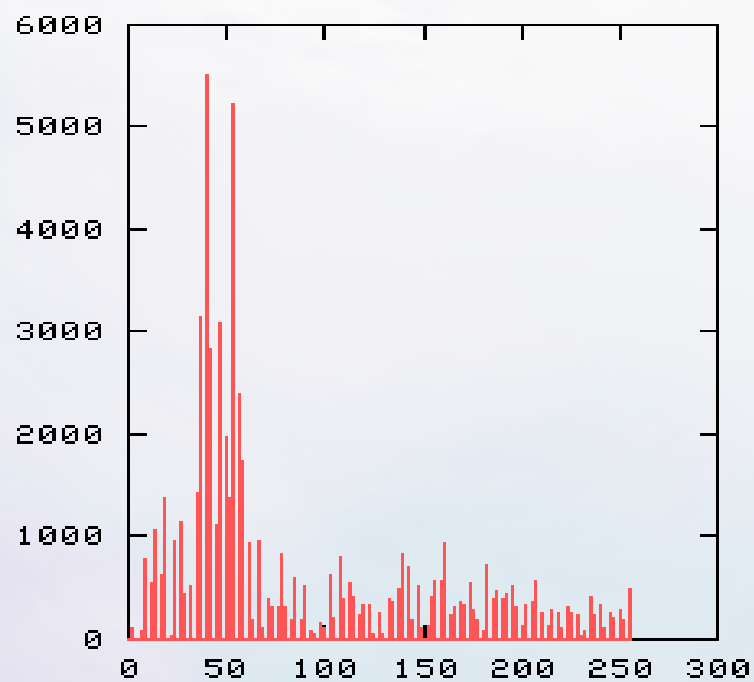
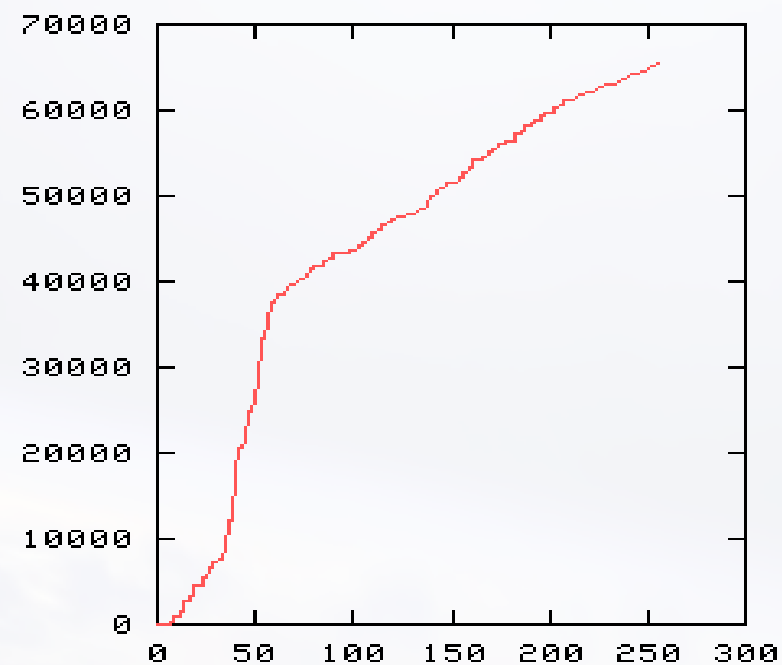
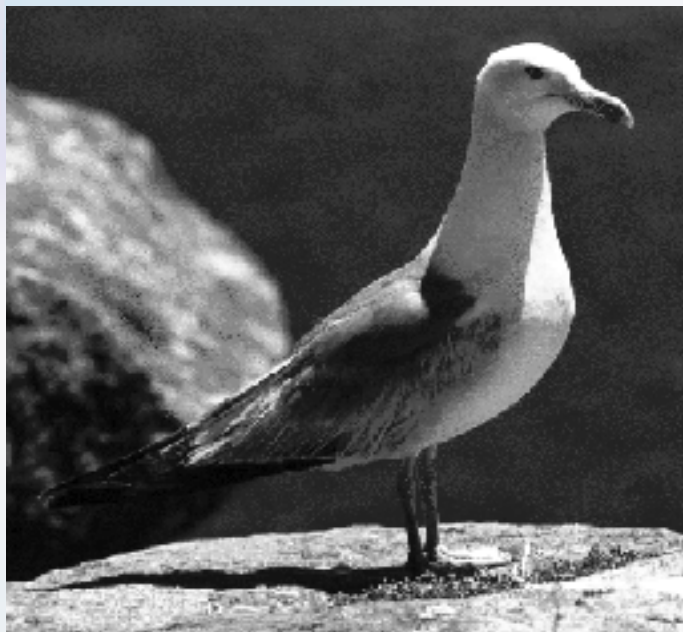
Histogram

- Kumulativni histogram slike sivih razina $H(k)$ (1D signal) predstavlja ukupan broj piksela (frekvenciju pojavljivanja) sa razinom sive manjom od iznosa k

$$H(k) = |\{f(x, y) : f(x, y) < k\}|$$

- Histogram slike sivih razina $h(k)$ (1D signal) predstavlja diskretnu razliku vrijednosti $H(k)$ i $H(k-1)$

$$h(k) = H(k) - H(k-1)$$



Histogram

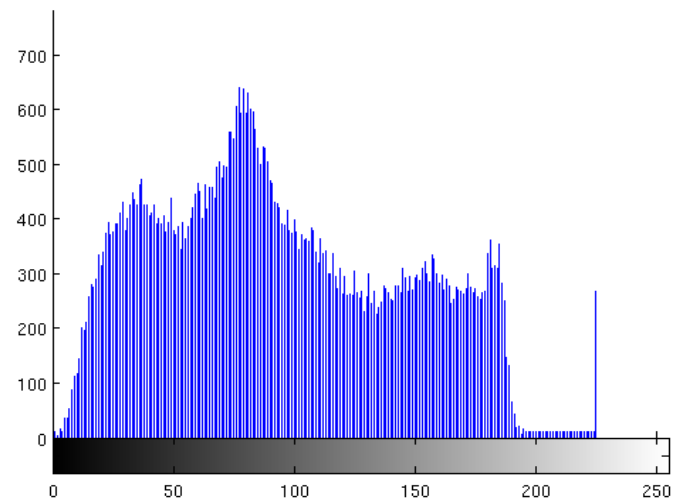
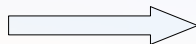
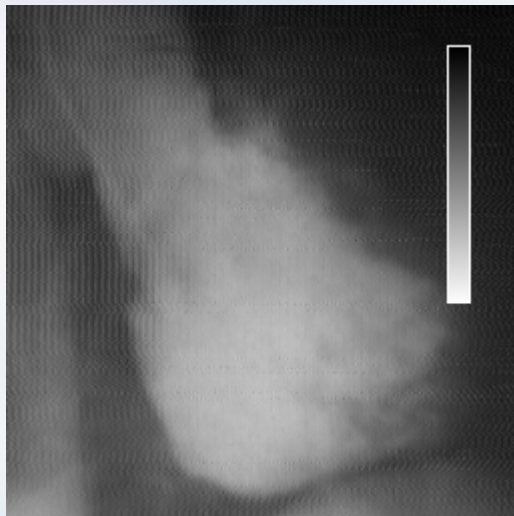
- Standardni način zapisa statističke distribucije frekvencija skupa ulaznih podataka
 - Diskretizacija domene
 - Prebrojavanje uzoraka koji poluliraju svaku ćeliju diskretne domene

U digitalnoj obradi slike:

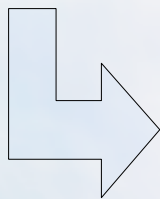
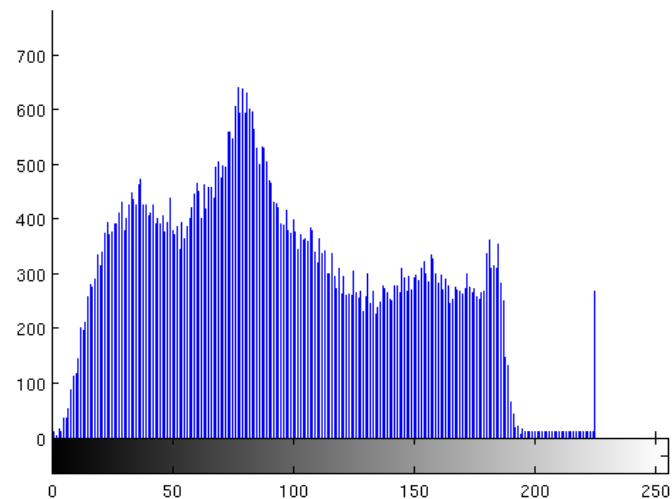
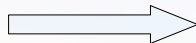
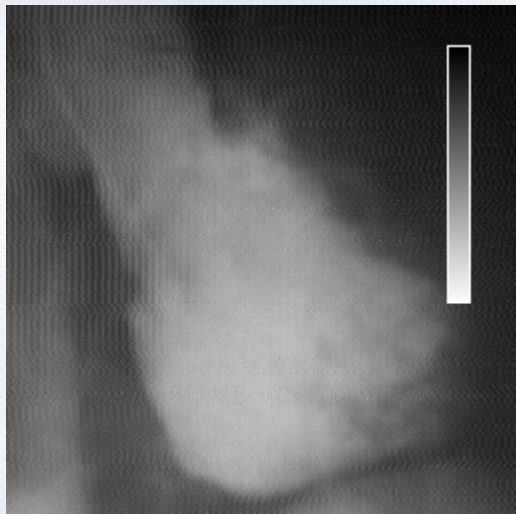
- Statistička distribucija frekvencija nijansi sive (1D signal)
- Statistička distribucija frekvencija za svaku od komponenti (kod slike u boji)
- Višedimenzionalni histogram: Statistička distribucija frekvencija u višedimenzionalnoj domeni

Izgled histograma ne ovisi o prostornoj distribuciji piksela na slici!

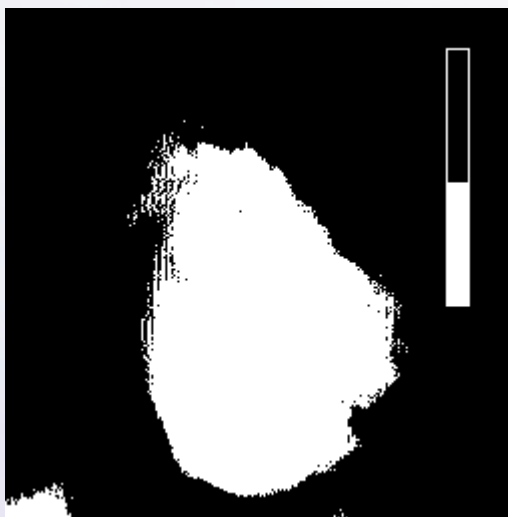
Amplitudna segmentacija korištenjem histograma



Amplitudna segmentacija korištenjem histograma

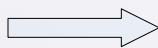
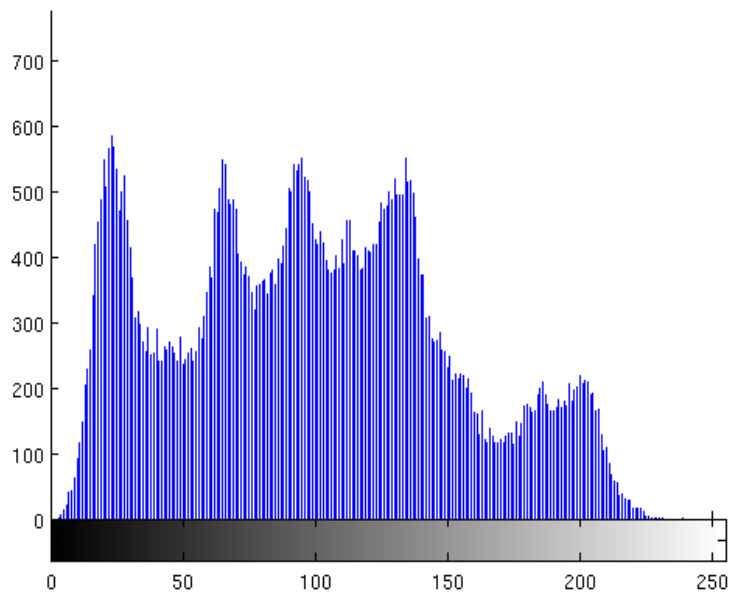
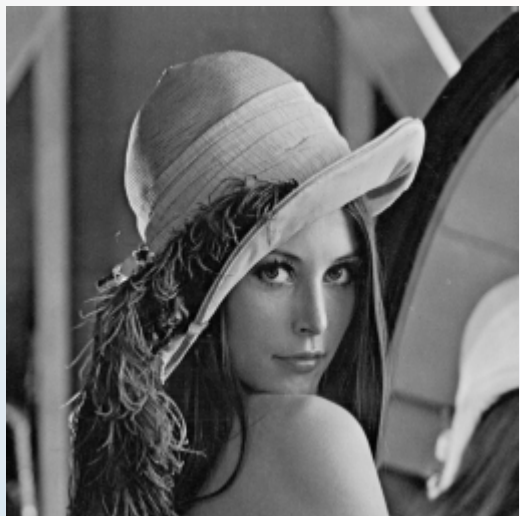


$T = 130$

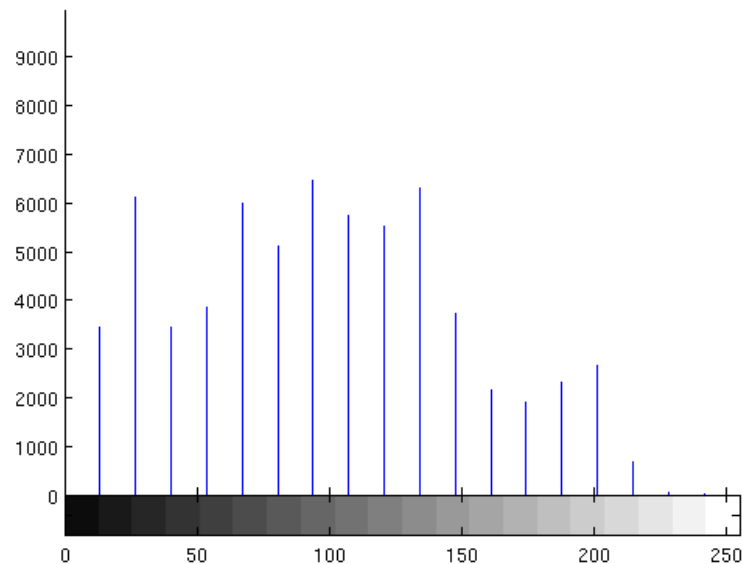


Histogram nijansi sive

imhist(A, 256)



imhist(A, 20)



Histogram RGB slike



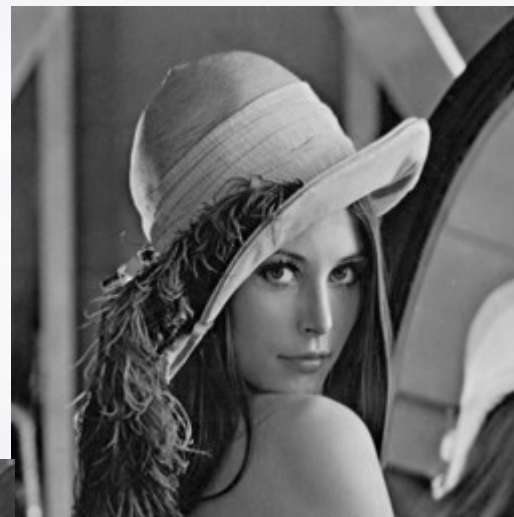
Histogram RGB slike



RED



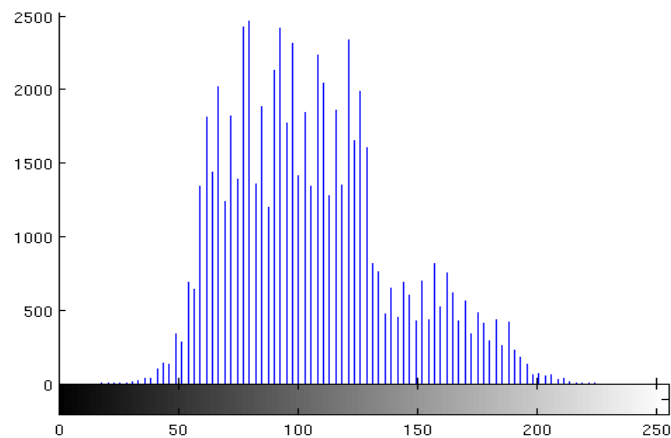
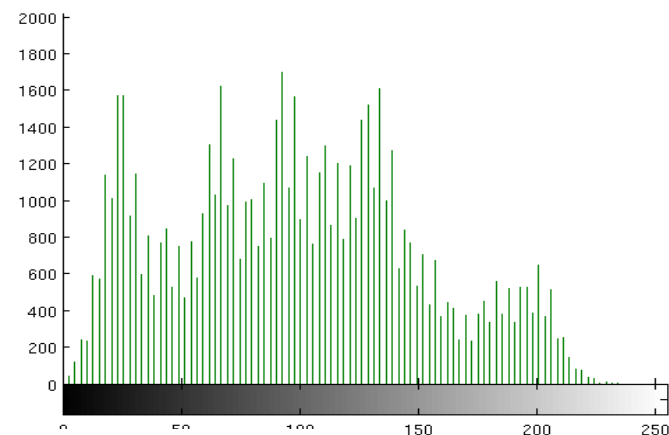
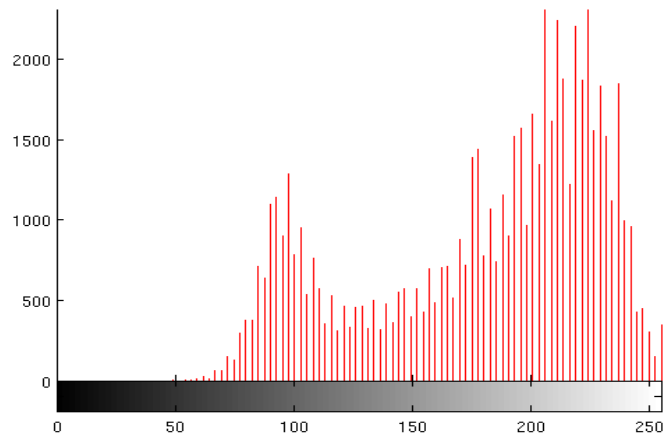
GREEN



BLUE

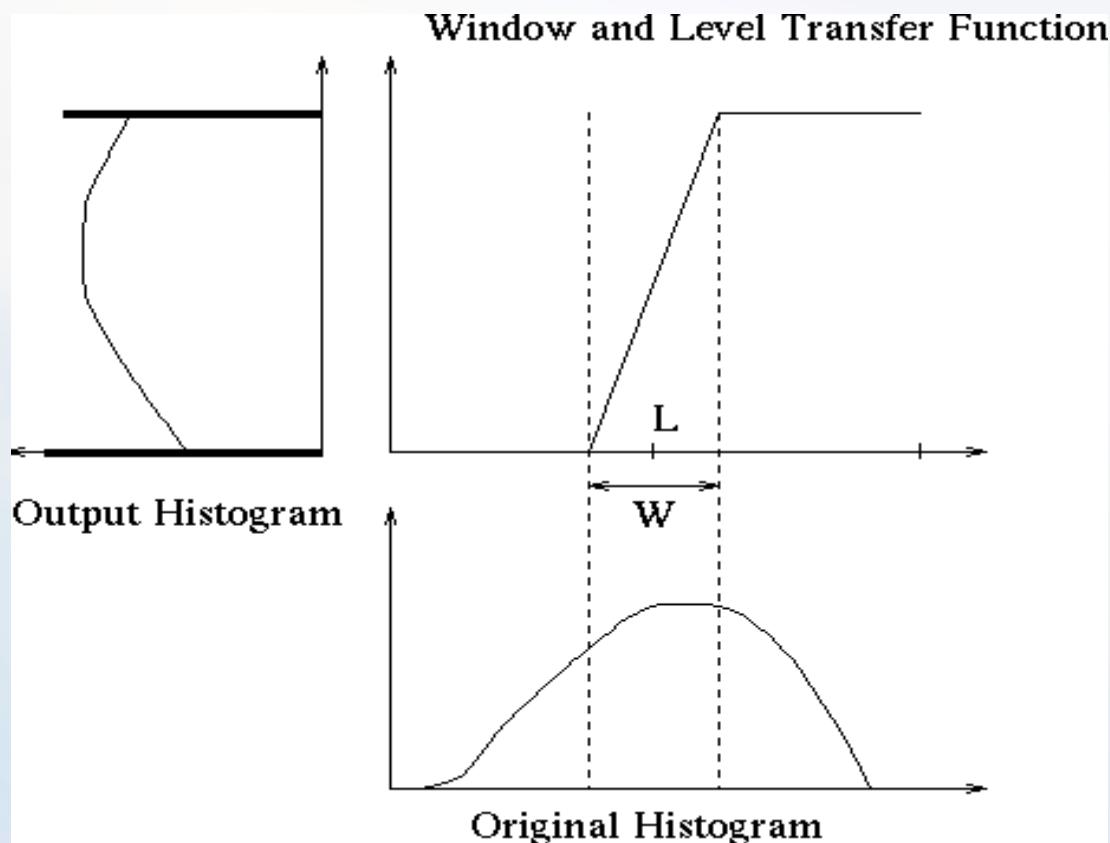


Histogram RGB slike

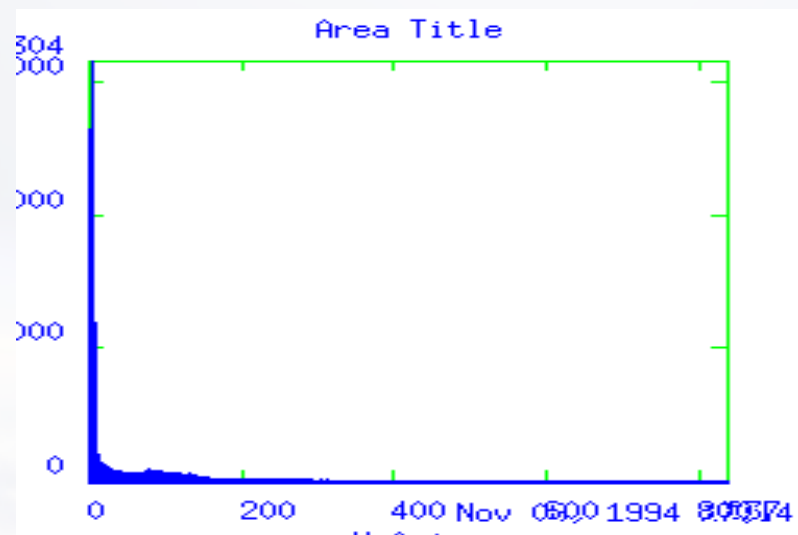
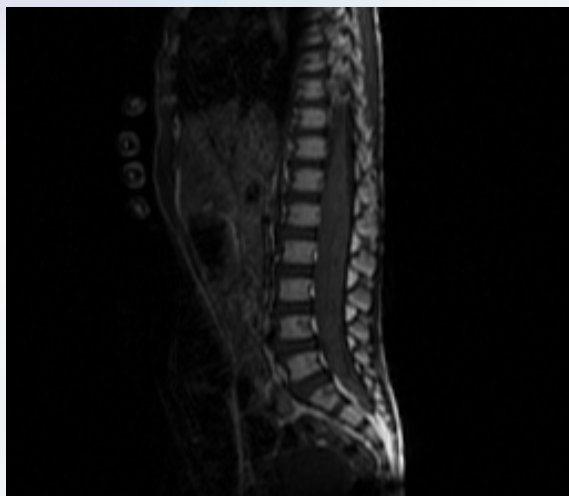


Kontrast

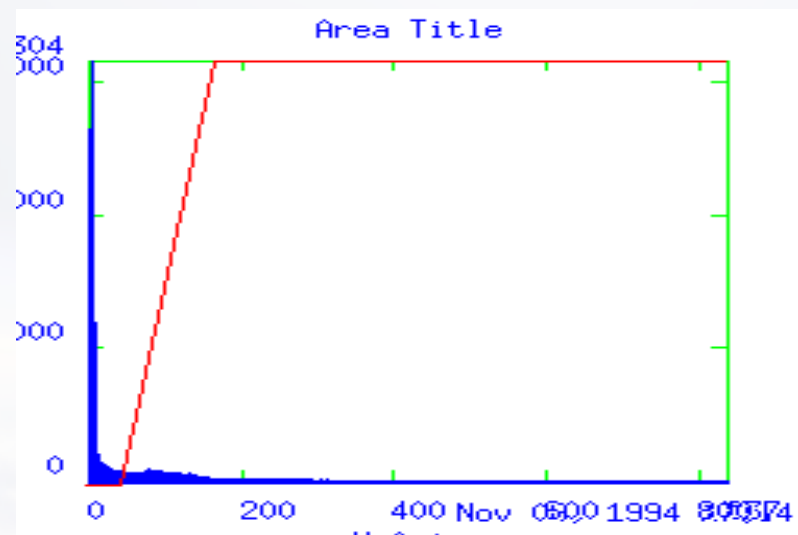
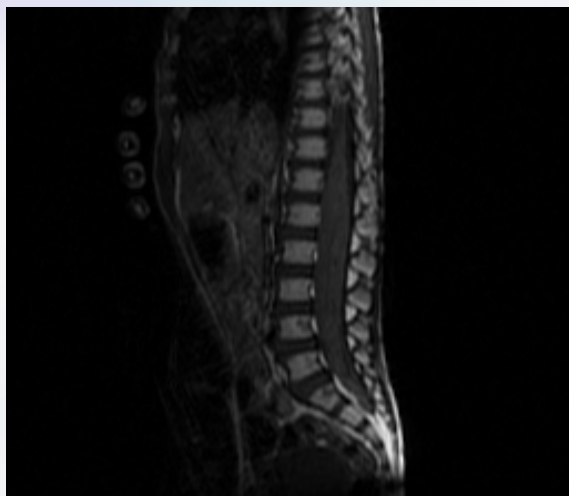
- **Intensity of Interest (IOI):** funkcijom preslikavanja ističe se ciljano područje histograma



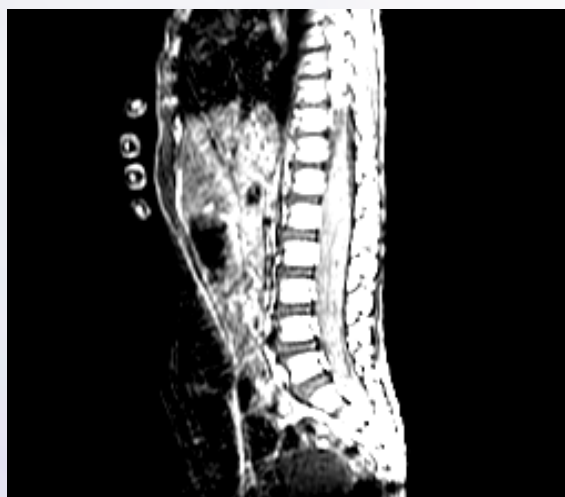
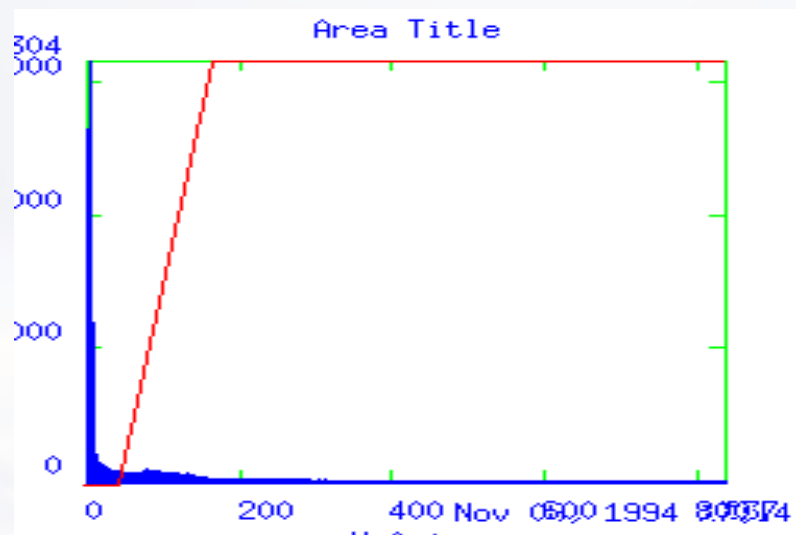
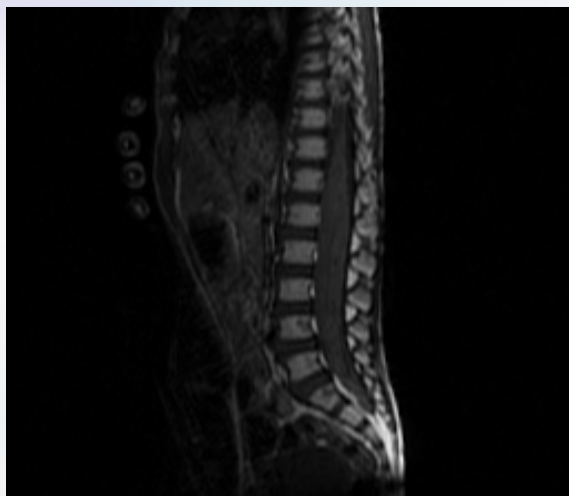
Poboljšanje kontrasta - primjer



Poboljšanje kontrasta - primjer



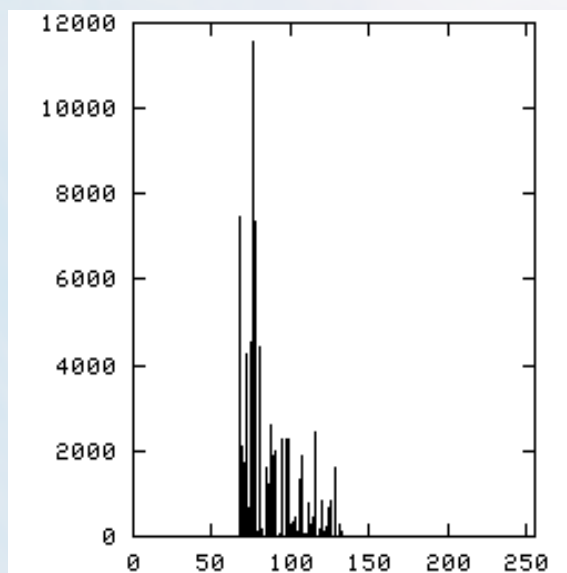
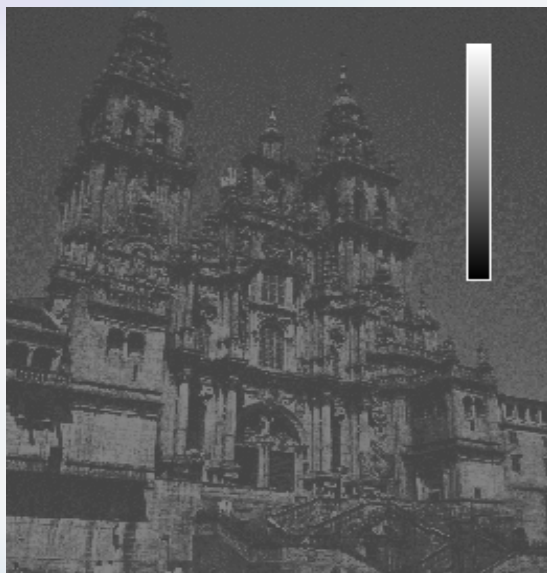
Poboljšanje kontrasta - primjer



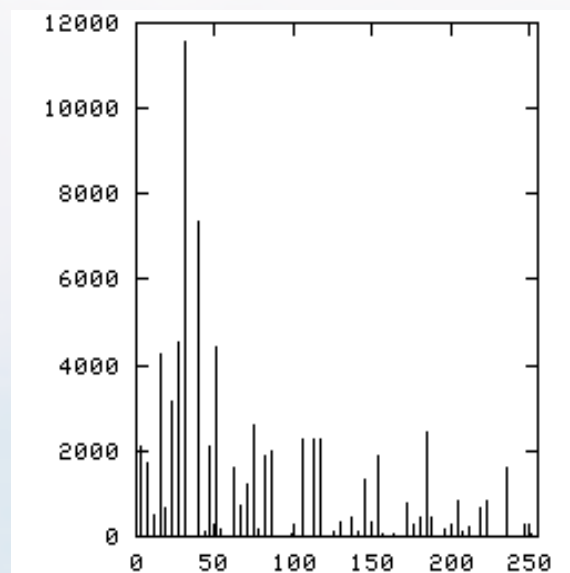
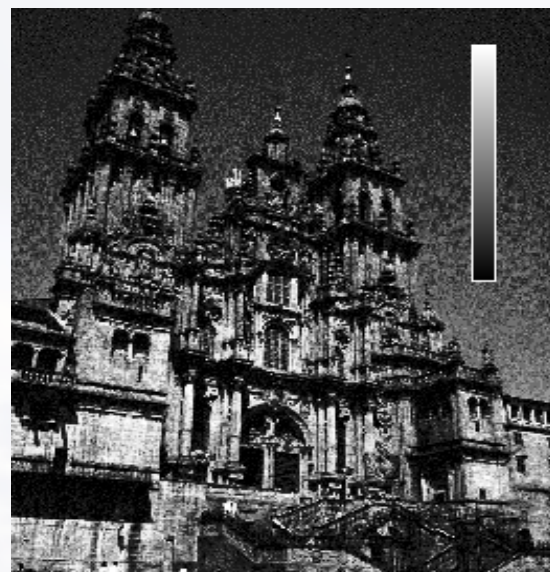
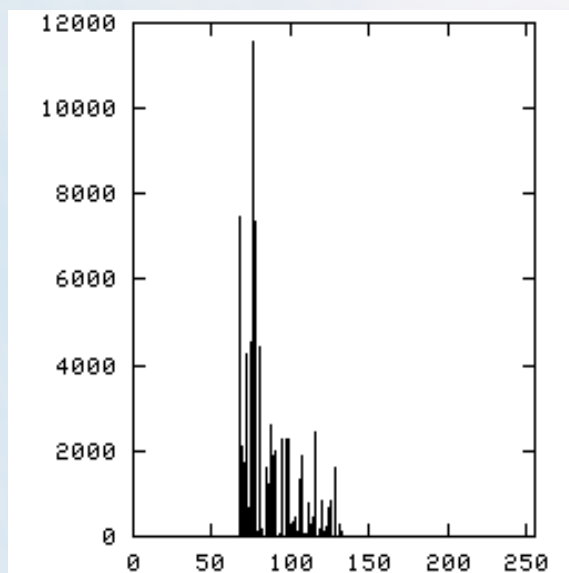
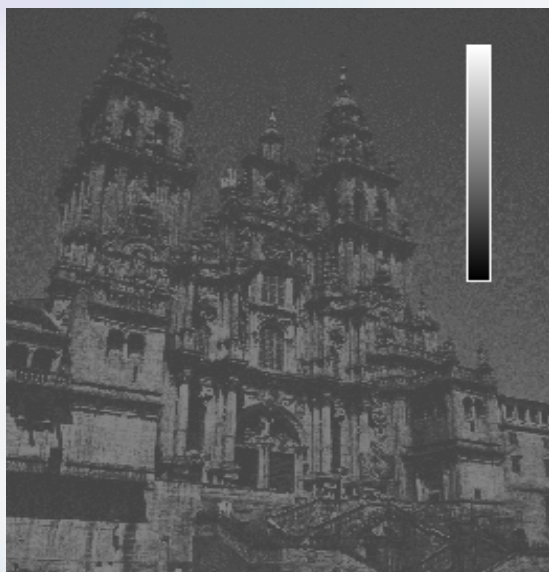
Rastezanje histograma

- **Histogram stretching:** povećanje kontrasta slike postavljanjem **IOI** područja na raspon svih vrijednosti piksela
- **Normalizacija – mapiranje minimalne vrijednosti na slici u 0, maksimalne vrijednosti u 255 (za 8 bitne slike)**
- Ova operacija je često implemetirana u automatici uređaja za prikaz slike

Rastezanje histograma - primjer



Rastezanje histograma - primjer



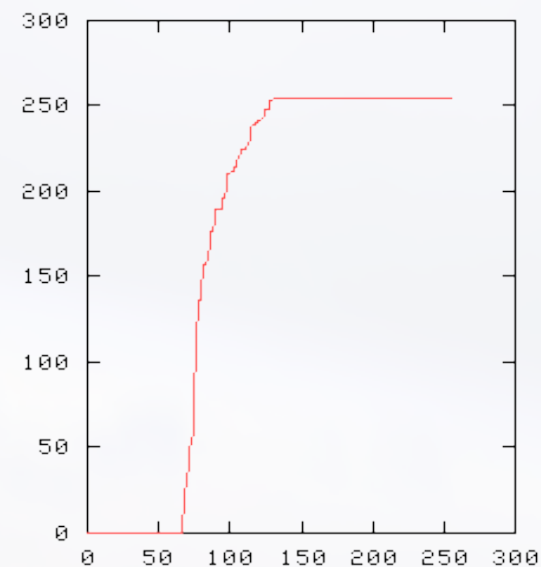
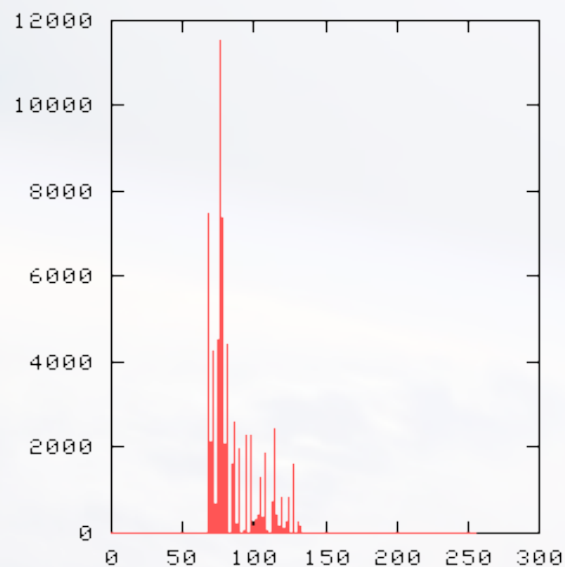
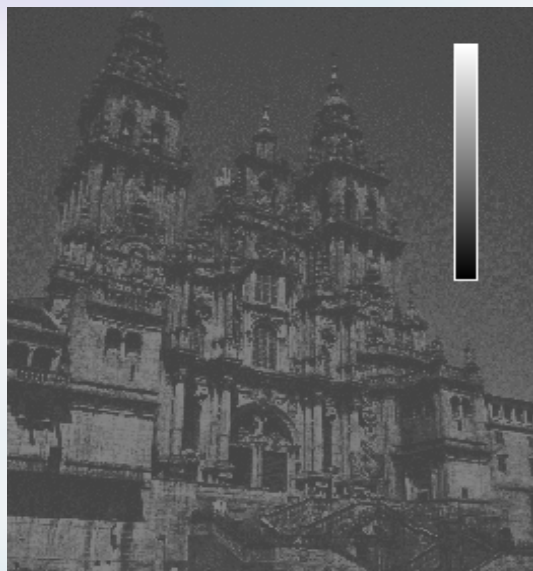
Ujednačavanje histograma

- **Histogram equalization**: tehnika povećanja kontrasta sa ciljem dobivanja ujednačenog histograma
 - Kao funkcija preslikavanja koristi se normalizirani kumulativni histogram

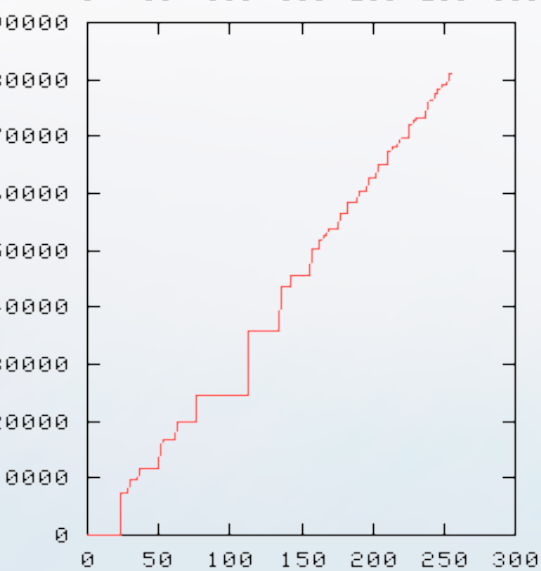
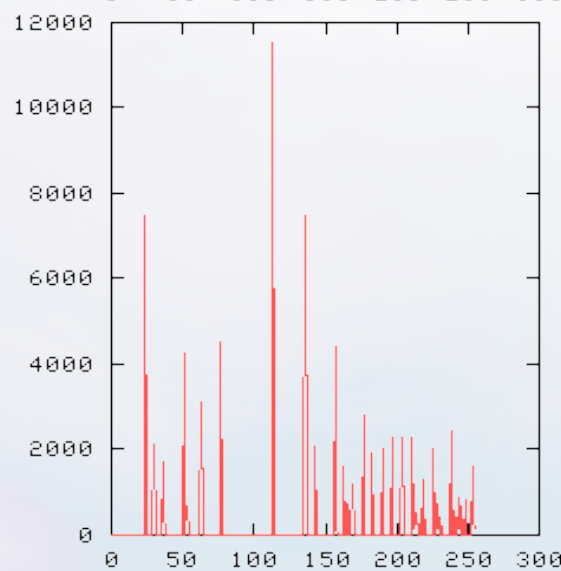
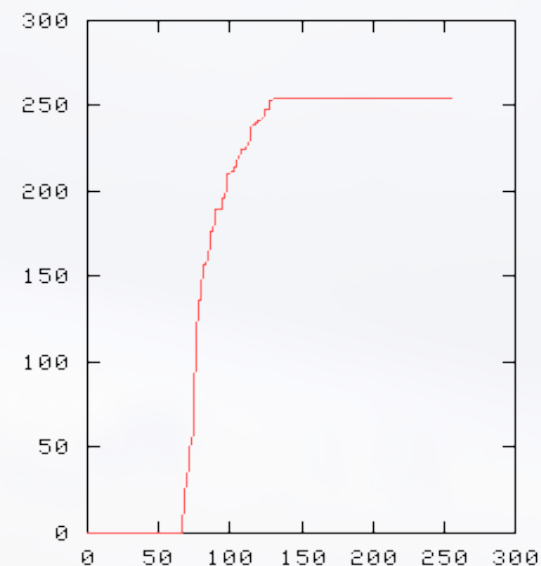
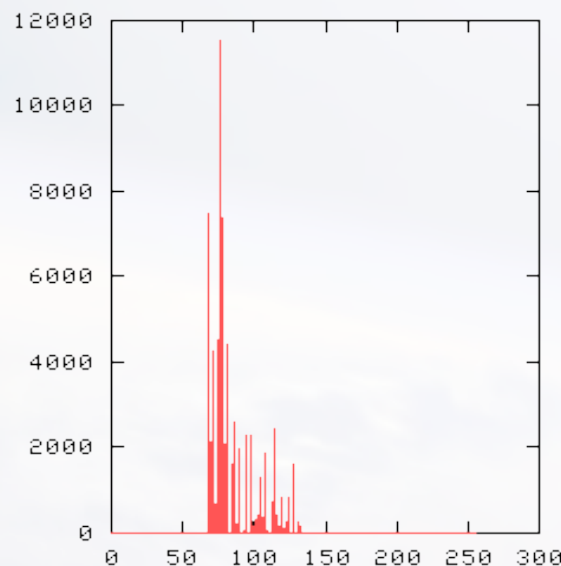
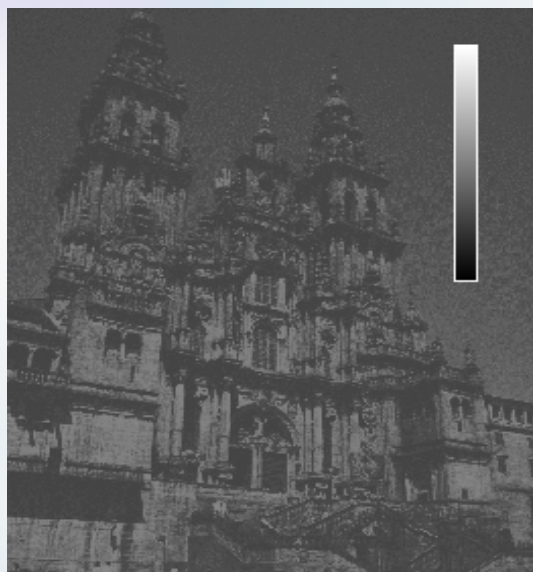
Histogram equalization

- (1) Računanje kumulativnog histograma ulazne slike
- (2) Normalizacija kumulativnog histograma na maksimalnu vrijednost piksela (255 za 8 bitne slike)
- (3) Preslikavanje korištenjem normaliziranog kumulativnog histograma kao funkcije preslikavanja

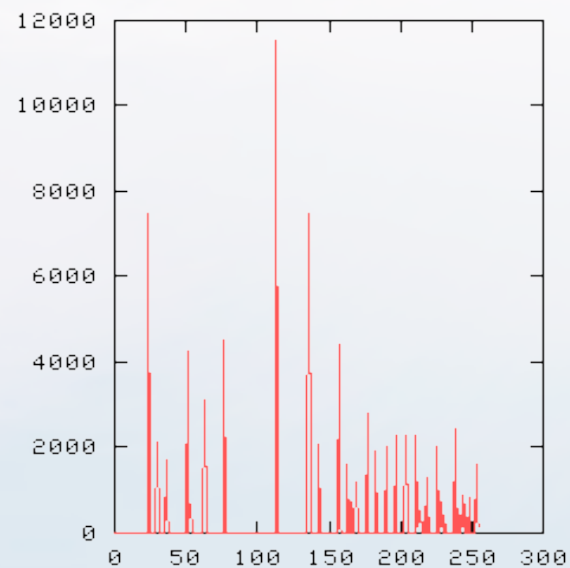
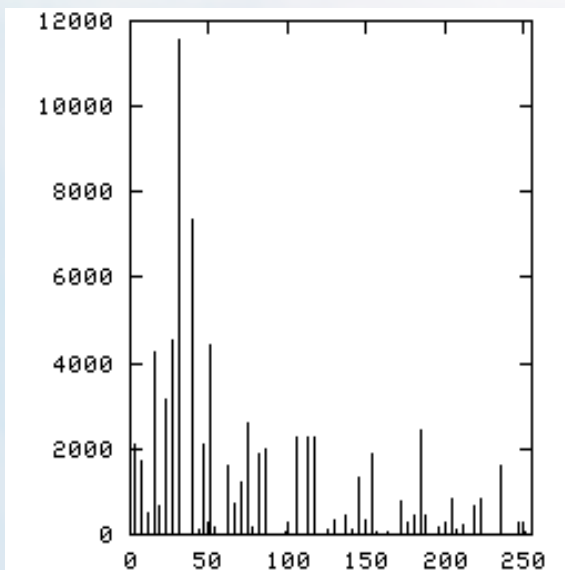
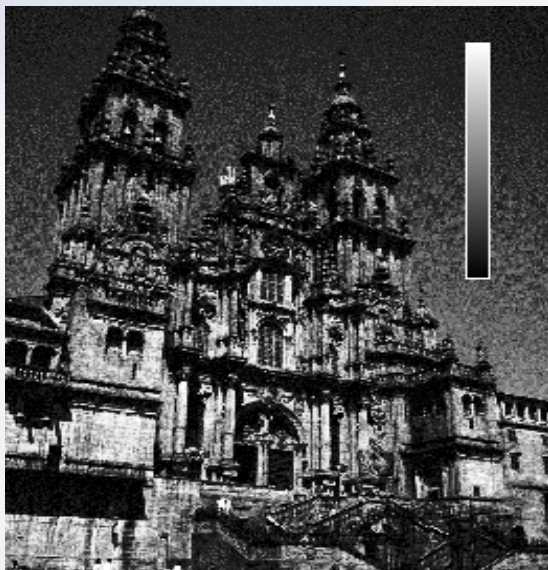
Ujednačavanje histograma - primjer



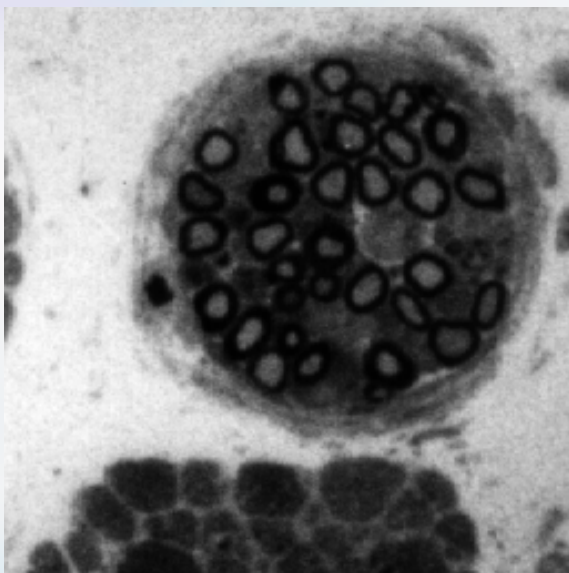
Ujednačavanje histograma - primjer



Stretching vs. Equalization

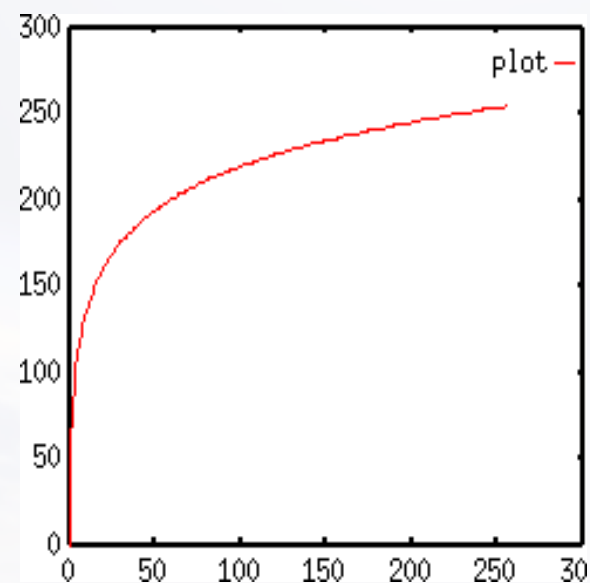
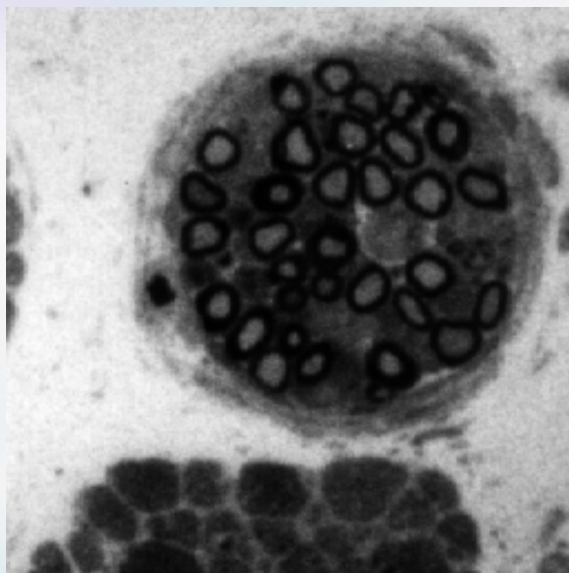


Zadatak:

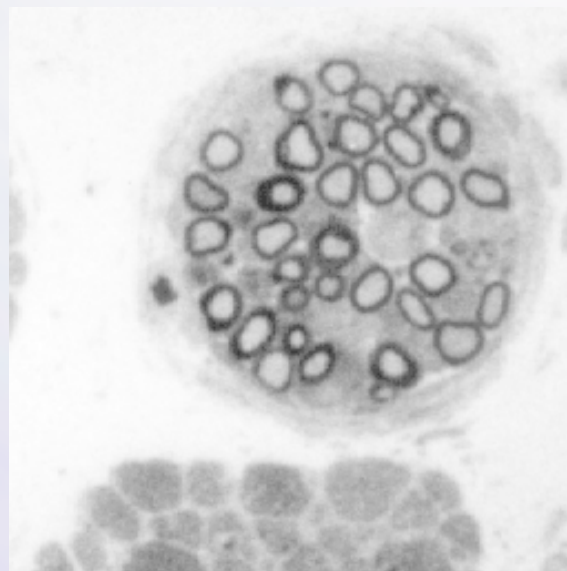
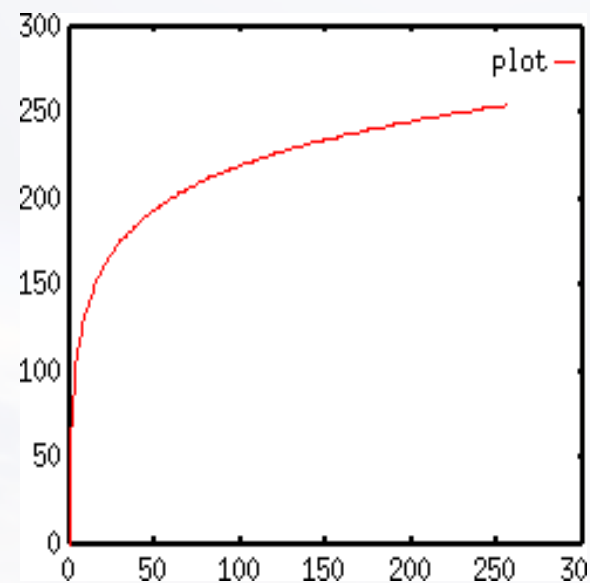
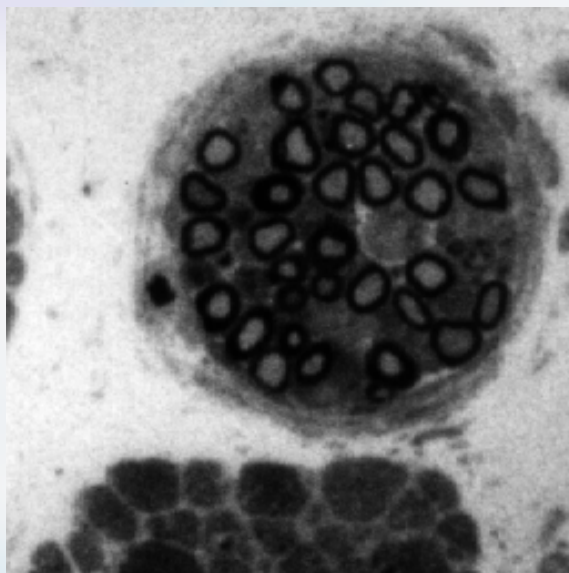


Na koji način bi bilo optimalno povećati kontrast ukoliko želite naglasiti detalje koji su u području tamnijih nijansi?

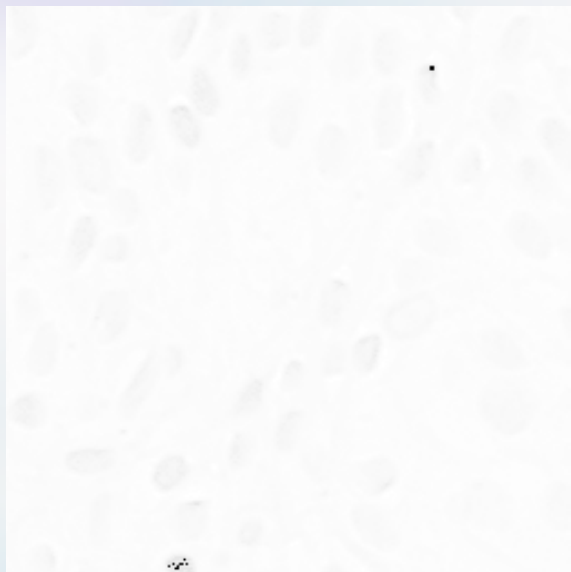
Logaritamska funkcija preslikavanja



Logaritamska funkcija preslikavanja

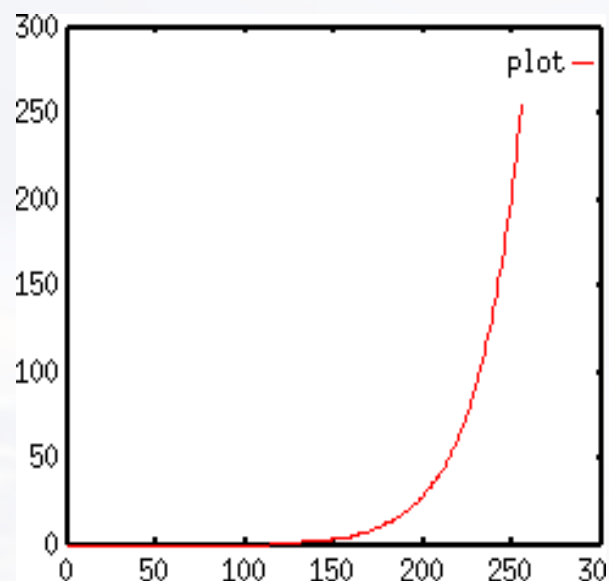
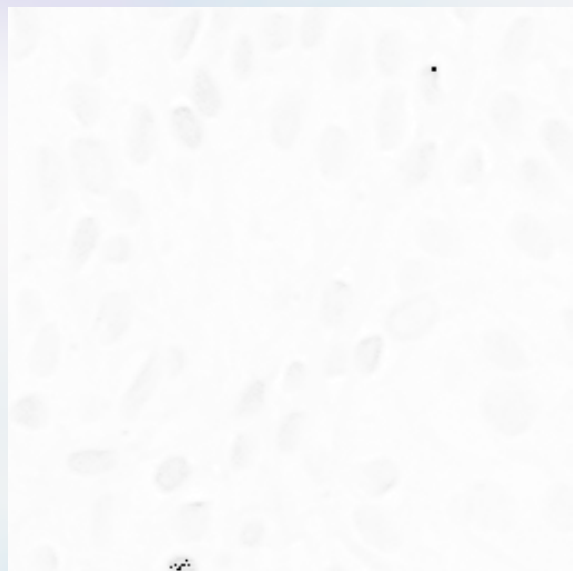


Zadatak:



Na koji način bi bilo optimalno povećati kontrast ukoliko želite naglasiti detalje koji su u području svjetlijih nijansi?

Eksponencijalna funkcija preslikavanja



Eksponencijalna funkcija preslikavanja

